A Graphical Model for Articulability:

Verifiable + Articulable + Taste, with pooled annotators

Notes for Alex's articulability project

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1 Setting

We have n items indexed by i, with observed input x_i (text) and observed quality judgment $y_i \in \{0, 1\}$ (or scalar). Judgments come from annotators indexed by a; depending on the dataset we may or may not see annotator identities.

We posit m *metrics* (rubric criteria) indexed by $j \in \{1, \ldots, m\}$. A metric is a named, in principle articulable, dimension of quality (e.g. "novelty", "statistical rigor"). The metric set itself can be observed (provided rubric) or latent (extracted from data). Metrics may be related hierarchically; let $\mathrm{pa}(j)$ denote the parents of metric j in a DAG (empty for flat models).

2 Generative model

For each metric j:

$$r_j \sim p(r) \qquad \text{rubric statement (observed if provided)} \tag{1}$$

$$\mathrm{ext}_j \sim p(\mathrm{ext} \mid r_j) \qquad \text{operationalization (regex / program / LLM judge)} \tag{2}$$

For each item i and metric j:

$$z_{i,j} \sim p(z \mid x_i, r_j, \{z_{i,k}\}_{k \in \mathrm{pa}(j)}) \qquad \text{latent "true" value of metric } j \text{ on } x_i \tag{3}$$

$$\hat{z}_{i,j} = \mathrm{ext}_j(x_i) + \varepsilon^{\mathrm{ext}}_{i,j} \qquad \text{measurable proxy, noise } \sigma_j^2 \tag{4}$$

For each annotator a (in the most general case):

$$b_a \sim \mathcal{N}(\bar{b}, \tau_b^2) \qquad \text{annotator threshold/intercept} \tag{5}$$

$$w_{a,j} \sim \mathcal{N}(\bar{w}_j, \tau_{w,j}^2) \qquad \text{personal weight on metric } j \tag{6}$$

$$\eta_{a,j}(\cdot) \sim p(\eta \mid r_j) \qquad \text{personal reading of metric } j \text{ (link function)} \tag{7}$$

For each item i:

$$s_i \sim \mathcal{N}(0, \sigma_s^2) \qquad \text{shared (community) inarticulable taste residual} \tag{8}$$

$$t_{i,a} \sim \mathcal{N}(0, \sigma_t^2) \qquad \text{personal inarticulable taste residual} \tag{9}$$

The judgment is generated by:

$$\mu_{i,a} = b_a \; + \; \sum_{j=1}^{m} w_{a,j} \, \eta_{a,j}(z_{i,j}) \; + \; s_i \; + \; t_{i,a} \tag{10}$$

$$y_{i,a} \sim \mathrm{Link}(\mu_{i,a}) \qquad \text{e.g. Bernoulli}(\sigma(\mu_{i,a})) \tag{11}$$

When rationales are observed (e.g. peer review), we additionally have

$$\rho_{i,a} \sim p(\rho \,|\, \{z_{i,j}\}_{j \in A_{i,a}}, \text{polarity}_{i,a,j}) \tag{12}$$

where $A_{i,a}$ is the (latent) set of metrics annotator a activated when judging x_i.

Decomposition into the three components.

$$\underbrace{y_{i,a}}_{\text{outcome}} \;=\; \underbrace{f(\hat{z}_{i,\cdot})}_{\text{verifiable}} \;+\; \underbrace{\sum_j w_{a,j}\big(\eta_{a,j}(z_{i,j}) - \hat{z}_{i,j}\big)}_{\text{articulable beyond verifiable}} \;+\; \underbrace{s_i + t_{i,a}}_{\text{taste}}$$

The verifiable part uses only the cheap proxies $\hat{z}$. The articulable part adds the information in z that $\hat{z}$ misses (the verifiability gap, weighted by importance). The taste part is the residual of human judgment that no metric—articulated or latent—explains.

3 Three regimes for annotator data

We rarely observe individual annotator parameters cleanly. The model degrades gracefully across three data regimes.

3.1 Regime A: crossed (multi-annotator-per-item)

Each item has multiple annotators, with overlap across items. *Identification is strongest here.* Examples in your portfolio: peer review (multiple reviewers per paper), code review (sometimes multiple reviewers per PR).

Joint likelihood:

$$p(\boldsymbol{y} \,|\, x, \theta) = \prod_{i,a} p\big(y_{i,a} \,|\, x_i, b_a, w_{a,\cdot}, \eta_{a,\cdot}, s_i, t_{i,a}\big).$$

Random-effects on annotators ($w_{a,j} \sim \mathcal{N}(\bar{w}_j, \tau^2_{w,j})$) let us estimate the population-mean weights $\bar{w}_j$ *and* their dispersion $\tau^2_{w,j}$. Cases where $\tau^2_{w,j} \approx 0$ identify community-shared metrics; large $\tau^2_{w,j}$ identifies metrics that mark personal preference.

3.2 Regime B: pooled / mixture (annotator identities unknown)

This is your most common case. You see y_i but not which annotator produced it, or you see "the community accepted this" as an aggregated label. Examples: notice-and- comment substantivity, creative-writing thread votes, joke-comment scores, patent acceptance, homepage newsworthiness.

Two sub-cases:

B1. Latent-mixture pool. A finite number K of annotator types with mixing weights $\boldsymbol{\pi}$. For each judgment we marginalize the type:

$$p(y_i \,|\, x_i) = \sum_{k=1}^{K} \pi_k \cdot p\big(y_i \,|\, x_i, b^{(k)}, w^{(k)}_{\cdot}, s_i\big).$$

$K = 1$ collapses to a single "community annotator"; $K > 1$ captures heterogeneity without needing IDs. Identifiable via EM if there is enough variability in x and the metric loadings differ across types.

B2. Marginalized-population pool. Treat each judgment as drawn from a fresh annotator with parameters from the population:

$$p(y_i \mid x_i) = \int p\big(y_i \mid x_i, b, w, s_i\big)\, \mathcal{N}(b; \bar{b}, \tau_b^2) \prod_j \mathcal{N}(w_j; \bar{w}_j, \tau_{w,j}^2)\, dw\, db.$$

With a logit link and Gaussian effects this is a standard random-effects logistic model; $\bar{w}$ and τ_w are identifiable from variance in y *conditional on* $\hat{z}$.

In both pooled cases:

- Personal taste $t_{i,a}$ and personal weight deviations $w_{a,j} - \bar{w}_j$ are *not separately* identified—they collapse into a single dispersion parameter τ_{pool}.
- Shared taste s_i is still identified, because it varies systematically with i across all annotators (it is the part of y predicted by x *after* regressing on z).
- The verifiability and articulability gaps remain identified (they are item-level quantities, not annotator-level).

3.3 Regime C: single consensus label

Only one y_i per item, no annotator information at all (e.g. patent grant decisions, many web-scraped labels). Equivalent to B2 with all annotator dispersion absorbed into noise. The decomposition reduces to:

$$y_i = f(\hat{z}_{i,\cdot}) + \sum_j \bar{w}_j \big(\eta_j(z_{i,j}) - \hat{z}_{i,j}\big) + s_i + \xi_i,$$

where ξ_i is irreducible noise that bundles annotator dispersion and personal taste. Cannot distinguish "community consensus residual" s_i from "noise" ξ_i unless you have repeated measurements per x_i or an external proxy.

4 The three gaps as restricted models

Fit nested predictors and read off the lifts:

$$\begin{aligned}
&\mathcal{M}_0: \quad \hat{y}_i = \beta_0 && && (13)\\
&\mathcal{M}_1: \quad \hat{y}_i = \beta_0 + f(\hat{z}_{i,\cdot}) && \text{verifiable only} && (14)\\
&\mathcal{M}_2: \quad \hat{y}_i = \beta_0 + \sum_j \bar{w}_j\, \eta_j(z_{i,j}) && \text{verifiable + articulable} && (15)\\
&\mathcal{M}_3: \quad \hat{y}_i = \text{dense}(x_i) && \text{everything (BT/regression on raw text)} && (16)
\end{aligned}$$

with metrics

$$\begin{aligned}
\text{Verifiability gap} &:= \text{AUC}(\mathcal{M}_2) - \text{AUC}(\mathcal{M}_1) && (17)\\
\text{Articulability gap} &:= \text{AUC}(\mathcal{M}_3) - \text{AUC}(\mathcal{M}_2) && (18)\\
\text{Taste residual variance} &:= \text{Var}\big(y_i - \mathcal{M}_3(x_i)\big) && (19)
\end{aligned}$$

In Regime A, further split the taste residual into shared vs. personal by ICC of residuals across annotators on the same item.

5 Identification conditions

1. Population means $\bar{w}_j$ and the verifiability gap require only x_i, y_i and a fixed set of $\hat{z}$ + LLM-judge z predictions. *Identifiable in all regimes.*
2. Annotator dispersion $\tau^2_{w,j}$ requires multiple judgments per item with systematic variation. *Regime A only.*
3. Shared taste s_i requires repeats or a sufficiently rich predictor space such that the dense model captures item-specific effects. *Regimes A and B.*
4. Personal taste $t_{i,a}$ requires both crossed annotators *and* sufficient coverage that we can attribute residuals to individuals. *Regime A only.*
5. Hierarchical structure on metrics (DAG via $\mathrm{pa}(j)$) requires either an observed rubric tree or extracted entailment relations between z_j's.

A quick heuristic: *which questions can each dataset answer?*

Dataset	V-gap	A-gap	Shared taste s_i	Personal taste $t_{i,a}$
Peer review (multi-reviewer + rationales)	✓	✓	✓	✓
Code review (multiple reviewers per PR)	✓	✓	✓	partial
Notice & comment (pooled)	✓	✓	✓	—
Creative writing (Reddit votes, pooled)	✓	✓	partial	—
Humor (joke score, pooled)	✓	✓	partial	—
Patents (single decision)	✓	✓	—	—
Press releases (single label)	✓	✓	—	—
Newsworthiness (homepage, pooled)	✓	✓	partial	—
Math (verifier ground truth)	✓	—	—	—

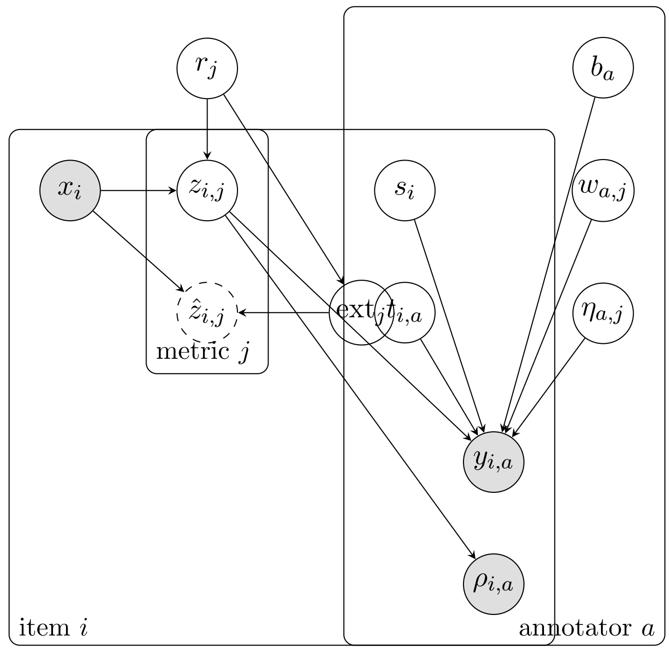
6 Choice of aggregation function

Linear-in-metrics (above) is convenient but is wrong in three common cases:

1. **Dealbreakers / vetoes.** A single fatal flaw rejects regardless of others (peer review, patents). Use noisy-AND on dealbreaker metrics or a min-pooling layer: $\mu = \min_{j \in D} z_{i,j} + \sum_{j \notin D} w_j z_{i,j}$.
2. **Threshold effects.** Quality jumps once a criterion crosses a level (humor, newsworthiness). Replace z with $\nVdash\{z > \theta_j\}$ before weighting.
3. **Substitutability.** Multiple ways to be good (creative writing: dense plot *or* strong voice). Use a softmax / log-sum-exp over substitutable metric bundles instead of a sum.

These are local refinements; the rest of the framework (gaps, identification) is unchanged.

7 Plate diagram



Shaded = observed; solid circle = latent; dashed = deterministic given parents. The annotator plate collapses to a single node in Regime C and to a marginalized random effect in Regime B.

8 Connection to your existing apparatus

- **Verifier-first programs** populate $\hat{z}_{\cdot,j}$ where ext_j is a synthesized program. Library convergence rate is your direct estimate of $\dim(\hat{z})$ for the task.
- **STaR / z_+, z_- extraction** produces samples from $p(\rho_{i,a} \mid z_{i,A_{i,a}}, \text{polarity})$, which is partial observation of the latent metric activations $A_{i,a}$.
- **Haupt entailment hierarchy** estimates the $\text{pa}(\cdot)$ DAG from the z's, allowing you to fit the hierarchical version of the model rather than the flat one.
- **Thin/thick classification** per metric is the empirical question of whether $\text{Var}(\hat{z}_{i,j} - z_{i,j}) \approx 0$ when ext_j is a program (thin) or requires an LLM (thick).

9 Open questions

1. Is $z_{i,j}$ a platonic latent (realist) or a community fixed-point (constructive)? Affects whether the model is generative or only descriptive.
2. In pooled regimes, does $K{=}1$ vs. $K{>}1$ in mixture pooling actually buy you anything, or does the population-effects formulation suffice?
3. Should aggregation be per-task (peer review uses dealbreakers; humor uses thresholds; creative writing uses substitution)? Probably yes; build a small library of aggregators and pick by held-out fit.

4. How do we handle the case where the rubric statement r_j is itself uncertain (extracted from text, not given)? Treat r_j as latent and infer with Dirichlet-process or fixed-vocabulary priors over rubric items.